

Surveying the Merits of Redesigning the OU Industrial & Systems Engineering Degree: A Cognitive Workload Approach

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Abstract

Cognitive workload has been shown to directly affect student burnout, well-being, and academic performance, and prior work on the OU Industrial & Systems Engineering (ISE) degree identified Junior-year Fall as the heaviest semester in the program. Building on that finding, this study evaluates whether three alternative degree layouts can redistribute perceived workload more evenly. Thirteen ISE upperclassmen completed a 28-item Qualtrics survey rating four layout-level metrics (difficulty, workload intensity, preference, anticipated graduation rate) and three semester-level metrics (workload intensity, anticipated failure rate, weekly hours worked) on visual-analog sliders adapted from NASA-TLX. A within-subjects design was used, with Shapiro-Wilk tests selecting between Friedman and ANOVA omnibus tests. At the semester level, three of three metrics differed significantly across the four semesters (workload intensity $p < 0.001$; failure rate $p < 0.001$; weekly hours $p = 0.002$), reaffirming Junior-Fall as the bottleneck. At the layout level, only one of four metrics differed significantly: anticipated graduation rate ($p = 0.023$). Difficulty, workload intensity, and preference were statistically indistinguishable across the four layouts. We interpret this asymmetry as evidence that participants recognize the bottleneck but resist redesigns that would resolve it—a pattern consistent with status-quo bias, anchoring, and the intrinsic-vs-extraneous load distinction in cognitive load theory. Limitations include the use of an outdated flowchart, the absence of counterbalancing, and the small response rate (13 of 50 invitations).

1 Introduction

1.1 Background and Motivation

Degree difficulty is, at its core, a cognitive workload problem. [Thornby et al. \(2023\)](#) define cognitive workload as the mental effort required to process, store, and manage task

information, and find that exceeding a student’s mental workload capacity diminishes not only performance, but also motivation and morale. [Zuo et al. \(2021\)](#) extend this finding by showing that calibrating curriculum intensity to students’ developmental abilities over time produces the best measurable outcomes.

Our previous work on the OU ISE curriculum identified a specific manifestation of this problem: Junior-year Fall consistently emerges as the most demanding semester, driven primarily by the co-placement of Data-Driven Decision Making I (DDDM I) and Deterministic Systems Models (DSM)—two of the three most difficult courses in the program. That finding established the existence of a bottleneck. The natural next question is whether the bottleneck can be relieved by redistributing courses without breaking the prerequisite chain. This study addresses that question by surveying ISE upperclassmen on three proposed alternative layouts (A, B, and C) alongside the current default, using a NASA-TLX-adapted instrument.

1.2 Literature Review

[Thornby et al. \(2023\)](#) document strong correlations between cognitive workload and institutional demands in health-professional education, and argue that workload measurement should inform curriculum design, teaching approaches, and student expectations. Although their setting is health-professional education, the underlying mechanism—a mismatch between cognitive capacity and external demand—generalizes across disciplines. Combined with [Zuo et al. \(2021\)](#), who found that workload calibrated to developmental readiness produces the highest cognitive engagement, the literature supports curricula whose difficulty scales in proportion to student abilities over time.

Education does not occur exclusively in physical settings. [Zuo et al. \(2021\)](#) examined cognitive workload across digital learning environments at elementary, middle, and high school levels and found a non-monotonic pattern: high-school students reported the heaviest academic workload alongside the lowest cognitive engagement, while elementary students reported the lightest workload and the highest engagement. The authors interpret this as evidence that manageable academic demand, paired with developmentally appropriate methods, maximizes engagement—and that simple video-call instruction may be insufficient to elicit the cognitive engagement seen in well-structured in-person settings.

Two design considerations from the broader literature shape the present instrument. First, [Krosnick \(1991\)](#) and [McDaniel and Gates \(2018\)](#) document that interspersing whimsical and humor-based items reduces satisficing behavior in long surveys; we adopt this practice in the engagement section of our instrument. Second, [Das Chakladar and Roy \(2024\)](#) review the established cognitive load taxonomy—intrinsic, extraneous, and germane—and apply it to physiological measurement of mental effort. The taxonomy will return in our discussion: a redesign that reduces *intrinsic* load (course-level difficulty) without addressing *extraneous* load (timing, sequencing, presentation) may not produce

the workload reduction students perceive as meaningful.

1.3 Research Question and Hypothesis

This study addresses two questions. (1) At the semester level, does the current ISE layout exhibit statistically significant differences in perceived workload intensity, failure rate, and weekly hours worked across the four core semesters (Sophomore-Fall, Sophomore-Spring, Junior-Fall, Junior-Spring)? (2) At the layout level, do the three proposed alternative degree layouts (A, B, C) differ from the current default in perceived difficulty, workload intensity, preference, or anticipated graduation rate? The formal null hypothesis for each variable is that all treatment levels share the same mean (or median, for non-normal data). The alternative hypothesis is that at least one treatment level differs from at least one other.

2 Methods

2.1 Participants

Participants were drawn from the current undergraduate junior and senior cohorts of the ISE program at the University of Oklahoma. This population was chosen because upper-classmen have completed enough of the curriculum to evaluate alternative layouts based on lived experience. Participation was voluntary and anonymous. Of 50 invitations distributed, 17 students initiated the survey and 13 provided complete responses; we proceed with $n = 13$ and treat the small sample as a limitation in Section 6.

2.2 Apparatus and Materials

The instrument was authored and distributed in Qualtrics. Excel was used for data cleaning, and R was used for all statistical tests and figures.

Quantitative measures. Perceived workload was elicited on visual-analog sliders. For all layout-level variables and the two semester-level rating variables (workload intensity, failure rate), sliders ranged from 0 to 100. For the semester-level weekly-hours-worked variable, the slider ranged from 0 to 40 to match the realistic range of a full-time courseload.

Qualitative measures. Three contextual sections framed the rating items: academic background (current standing, prior major), external influences (financial, work, family commitments per [Thornby et al. \(2023\)](#)), and engagement (humor-based items per [Krosnick \(1991\)](#); [McDaniel and Gates \(2018\)](#)).

2.3 Variables and Treatment Levels

Four layout-level variables were rated across four layout treatment levels each, and three semester-level variables were rated across four semester treatment levels each, for $4 \times 4 + 3 \times 4 = 28$ rating scenarios (Table 1). Each scenario was estimated to take roughly 20 seconds to complete.

Table 1: Variables and Treatment Levels

Genre	Variables (DV)	Treatment Levels (IV)
Layout	Difficulty; Workload Intensity; Preference; Anticipated Graduation Rate	Current (Default); Layout A; Layout B; Layout C
Semester	Workload Intensity; Anticipated Failure Rate; Weekly Hours Worked	Sophomore-Fall; Sophomore-Spring; Junior-Fall; Junior-Spring

All rating sliders run 0–100 except Weekly Hours Worked, which runs 0–40. Subjective measures were chosen because (i) they reflect lived student experience, (ii) they enable estimates of unimplemented future states, and (iii) they align with NASA-TLX recommendations for measuring subjective experiences.

2.4 Layout Designs

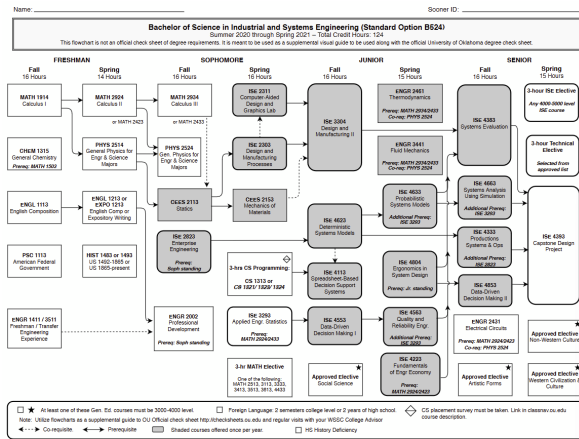
The three alternative layouts were designed to relieve Junior-Fall by separating DDDM I and DSM, two of the three highest-difficulty courses identified in our prior work. **Layout A** swaps DSM (currently Junior-Fall) with Probabilistic Systems Models (PSM, currently Junior-Spring). **Layout B** shifts both Applied Engineering Statistics (AES) and DDDM I back one semester each. **Layout C** combines both modifications. The four flowcharts (Default, A, B, C) appear in Figure 1; differences between each alternative and the default are highlighted in red and green to aid recognition.

2.5 Empirical Framework

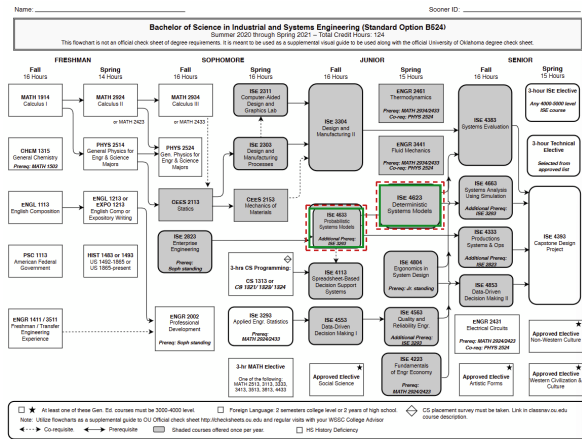
A within-subjects design was used. Due to Qualtrics limitations, counterbalancing was not implemented (a limitation revisited in Section 6). For each layout variable $k \in \{1, \dots, 4\}$ and each semester variable $k \in \{1, 2, 3\}$, the rating from participant i on treatment level j is modeled as

$$Y_{ij}^{(k)} = \mu^{(k)} + \tau_i^{(k)} + \varepsilon_{ij}^{(k)}, \quad i \in \{1, \dots, 13\}, \quad j \in \{1, 2, 3, 4\}, \quad (1)$$

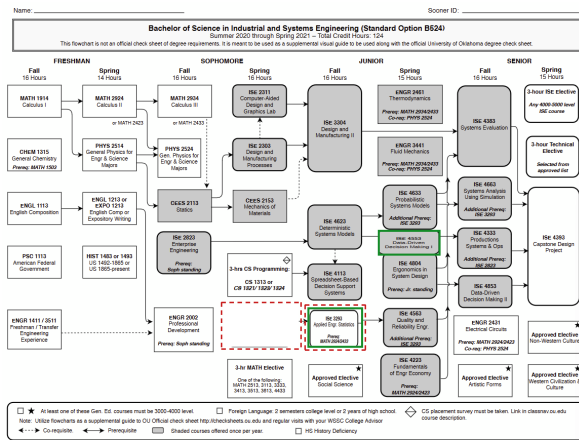
where $\mu^{(k)}$ is the grand mean for variable k , $\tau_i^{(k)}$ is the participant random effect, and $\varepsilon_{ij}^{(k)}$ is unattributable error.



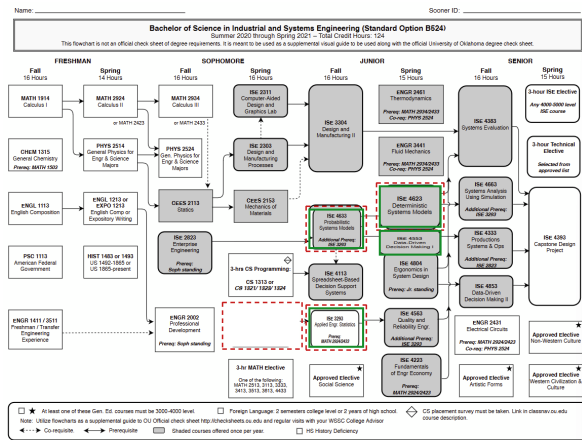
(a) Default Layout (Control)



(b) Layout A: Swap DSM & PSM



(c) Layout B: AES, DDDM I back one semester



(d) Layout C: Layouts A and B combined

Figure 1: Four degree-layout treatment levels. Differences from the default are highlighted in red (course moved *out*) and green (course moved *in*).

2.6 Statistical Tests

The four-stage testing pipeline (Figure 2) was applied to each of the seven dependent variables independently:

1. Apply Shapiro-Wilk to each of the four treatment levels of the variable.
2. If all four treatment levels are normally distributed ($p > 0.05$ in every cell), apply ANOVA to test equality of means.
3. If any treatment level rejects normality ($p < 0.05$ in any cell), apply Friedman’s test to test equality of medians.
4. Compare the omnibus p -value to $\alpha = 0.05$.

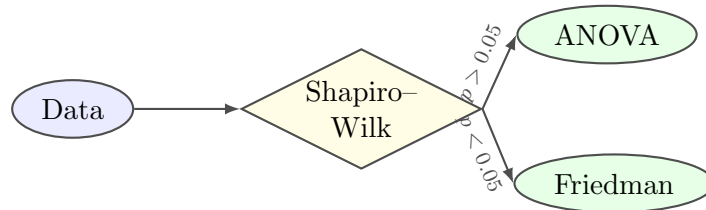


Figure 2: Statistical testing pipeline. Each of the seven dependent variables is independently routed to ANOVA or Friedman based on whether all four of its treatment levels pass Shapiro-Wilk.

3 Results

3.1 Descriptive Statistics

The layout boxplots (Figure 3) show that medians for difficulty, workload, and preference are visually similar across the four layouts, with the Default scoring slightly higher on difficulty and workload and Layout C scoring slightly higher on preference. The semester boxplots (Figure 4) show a much sharper pattern: Junior-Fall sits visibly above the other three semesters on all three metrics, and Junior-Spring sits second.

3.2 Normality and Test Selection

Five of the 28 treatment-level cells reject normality (Table 2); the remaining 23 are reported in Appendix A.

3.3 Omnibus Tests

The asymmetry in Table 3 is the central finding. **All three** semester-level variables differ significantly across the four semesters of the current layout: workload intensity ($p < 0.001$), anticipated failure rate ($p < 0.001$), and weekly hours worked ($p = 0.002$). The boxplots in Figure 4 make clear that Junior-Fall is the driver in each case. By contrast, **only one of**

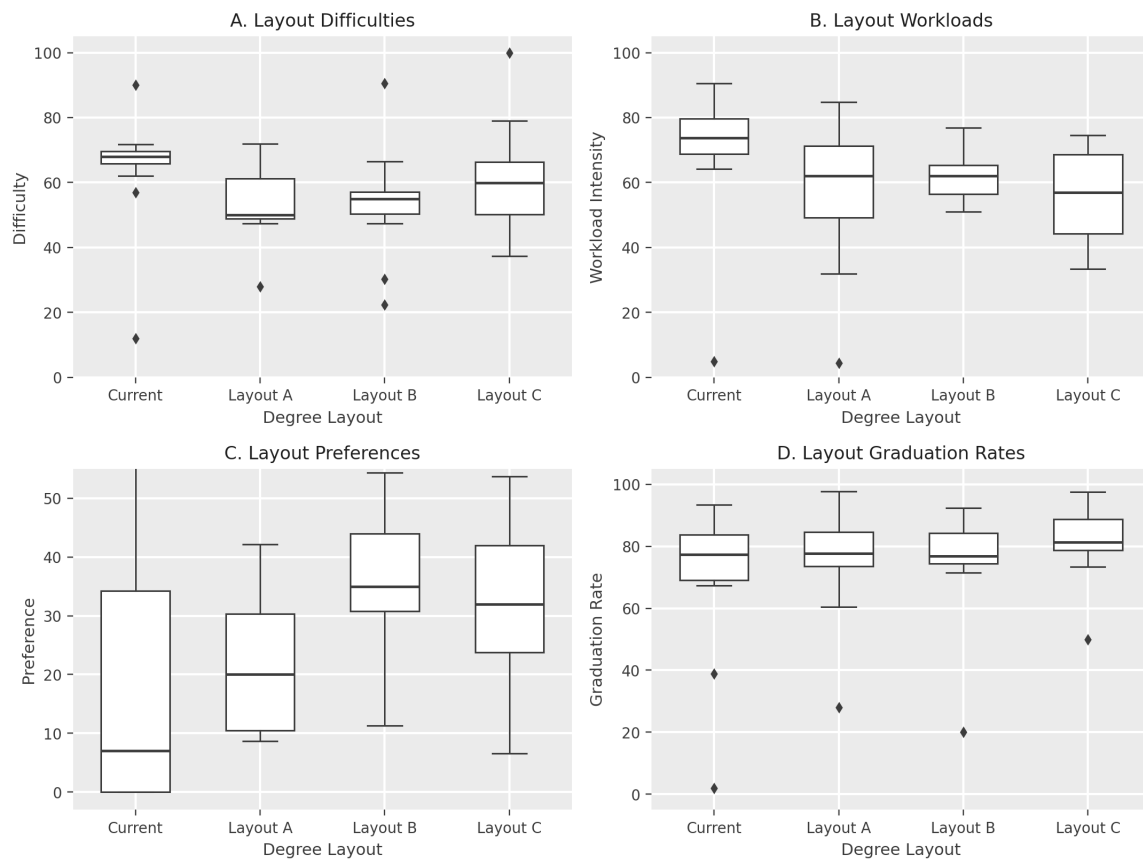


Figure 3: Boxplots of layout-level ratings (n=13). Panel A: Difficulty. Panel B: Workload Intensity. Panel C: Preference. Panel D: Anticipated Graduation Rate.

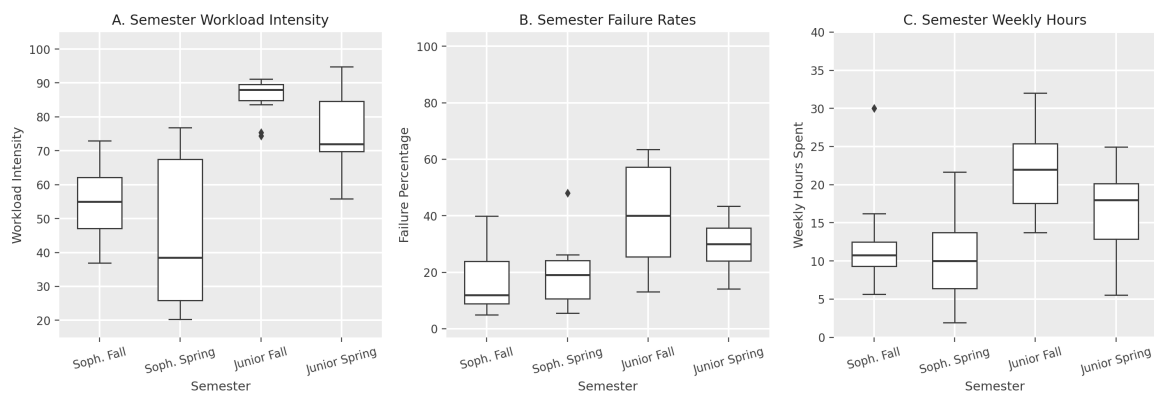


Figure 4: Boxplots of semester-level ratings (n=13). Panel A: Workload Intensity. Panel B: Anticipated Failure Rate. Panel C: Weekly Hours Worked.

Table 2: Treatment Levels Rejecting Normality (Shapiro-Wilk, $p < 0.05$)

Variable	Treatment Level	Shapiro-Wilk p
Layout Difficulty	Current (Default)	0.008
Layout Graduation Rate	Layout A	0.030
Layout Graduation Rate	Layout B	0.001
Semester Workload Intensity	Junior-Fall	0.003
Semester Failure Rate	Sophomore-Spring	0.010

Five of 28 treatment-level cells reject Shapiro-Wilk normality at $\alpha = 0.05$. Per the pipeline in Figure 2, the entire variable containing any rejecting cell is routed to Friedman’s test.

Table 3: Omnibus Test Results by Variable

Variable	Test	p -value	Significant ($p < 0.05$)
Layout Difficulty	Friedman	0.207	No
Layout Workload Intensity	ANOVA	0.704	No
Layout Preference	ANOVA	0.146	No
Layout Graduation Rate	Friedman	0.023	Yes
Semester Workload Intensity	Friedman	< 0.001	Yes
Semester Failure Rate	Friedman	< 0.001	Yes
Semester Weekly Hours Worked	ANOVA	0.002	Yes

Test selection follows the pipeline in Figure 2. All three semester-level variables reject equality of treatment levels; only one of four layout-level variables does.

four layout-level variables—anticipated graduation rate ($p = 0.023$)—differs significantly across the four layouts. The graduation-rate result is itself driven substantially by one outlying respondent who rated the current Layout, A, and B all below 25 (visible as low fliers in Figure 3D); the difficulty, workload, and preference variables are statistically indistinguishable across layouts at $\alpha = 0.05$.

In short: students confirm Junior-Fall is the bottleneck, but they do not perceive the three proposed redesigns as meaningfully different from the default—or from each other—on difficulty, workload, or preference.

4 Discussion

The semester-level results replicate and extend our prior finding that Junior-Fall is the heaviest semester in the OU ISE curriculum. Across all three semester-level metrics, Junior-Fall sits above the other three semesters at $p < 0.01$ or better, and Junior-Spring sits second. This is consistent with [Thornby et al. \(2023\)](#): when external demand exceeds cognitive capacity for a sustained period, students experience measurable workload intensity, failure risk, and time pressure simultaneously.

The layout-level results are harder to read at face value. Three of four layout variables fail to distinguish the alternatives from the default. The straightforward interpretation is

that participants do not perceive Layouts A, B, and C as meaningful improvements. There are at least three plausible mechanisms behind this pattern, and they are not mutually exclusive.

Status-quo and anchoring bias. The Default layout is shown first and serves as a reference point for every subsequent rating. Students may anchor on the Default’s perceived difficulty and rate the alternatives as roughly comparable simply because the alternatives *look* similar at flowchart granularity, even when the courses moved are structurally significant. Status-quo bias would compound this: participants may prefer the familiar default over alternatives, even alternatives specifically designed to address a problem they have themselves identified.

Intrinsic vs. extraneous load. Drawing on the cognitive load taxonomy reviewed in [Das Chakladar and Roy \(2024\)](#), our redesigns target intrinsic load—the inherent difficulty of co-placed courses—without explicitly addressing extraneous load (sequencing of prerequisites, timing of assessments, presentation of material). If students implicitly distinguish the two, they may correctly perceive that moving DDDM I or DSM by one semester does not reduce the underlying intrinsic difficulty of those courses, only redistributes when students encounter them. [Zuo et al. \(2021\)](#) support this reading: workload reduction without developmental scaffolding does not produce engagement gains.

Genuine equivalence. It is also possible the layouts genuinely are not very different from the default in expected workload, because moving the bottleneck to an earlier semester simply shifts the workload peak rather than flattening it. Layout B, in particular, moves AES and DDDM I back into Sophomore-Spring—which is currently the lightest semester—and so converts the Junior-Fall peak into a Sophomore-Spring peak rather than eliminating it.

The three mechanisms are observationally similar in this dataset and we cannot distinguish among them with $n = 13$ and no counterbalancing. What we *can* say is that the gap between students’ clear identification of Junior-Fall as the problem and their indistinguishable evaluation of the proposed solutions is the central paradox of this study.

5 Limitations and Future Research

Outdated flowchart. The four layouts shown to participants used the legacy ISE flowchart (Standard Option B524, Summer 2020–Spring 2021). The current curriculum has since added Systems Thinking and removed Physics for Engineers II, Electrical Circuits, Thermodynamics, and Fluid Mechanics. Replicating this study against the current flowchart is the single highest-priority extension.

No counterbalancing. Qualtrics limitations prevented question-order randomization, so all participants saw the four layouts in the same order (Default, A, B, C). Counterbalancing would directly mitigate the anchoring concern raised in Section 4.

Small sample. 13 of 50 invitations resulted in complete responses ($\approx 26\%$ completion rate, with another 4 partial responses excluded). With $n = 13$, the Friedman tests have limited power to detect small layout-level differences, and the marginal layout-graduation-rate result ($p = 0.023$) is sensitive to single outliers.

Modified rather than validated NASA-TLX. The instrument adapted NASA-TLX dimensions to fit a curriculum-design context rather than administering the validated instrument directly. Future work should administer the full NASA-TLX alongside the layout-rating items so that the two can be compared.

Single institution. Replicating across multiple ISE programs would clarify whether the bottleneck pattern and the resistance-to-redesign pattern are local to OU or generalize.

We conclude that the proposed layouts cannot be statistically distinguished from the default on difficulty, workload, or preference at this sample size, but that future work with the current flowchart, counterbalancing, and a larger sample may yet refute this preliminary result.

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Appendix A. Treatment Levels Passing Normality

Appendix B. Survey Instrument

The 28-item Qualtrics survey was administered in the following order:

1. *Background:* Current standing (Freshman / Sophomore / Junior / Senior / Other); self-rated competence in ISE (0–100 slider); prior major (free text).

Table 4: Treatment Levels Failing to Reject Normality

Variable	Treatment Level	Shapiro-Wilk p
Layout Difficulty	Layout A	0.259
Layout Difficulty	Layout B	0.999
Layout Difficulty	Layout C	0.604
Layout Workload Intensity	Current	0.076
Layout Workload Intensity	Layout A	0.583
Layout Workload Intensity	Layout B	0.403
Layout Workload Intensity	Layout C	0.796
Layout Preference	Current	0.118
Layout Preference	Layout A	0.353
Layout Preference	Layout B	0.434
Layout Preference	Layout C	0.313
Layout Graduation Rate	Current	0.108
Layout Graduation Rate	Layout C	0.358
Semester Workload Intensity	Sophomore-Fall	0.283
Semester Workload Intensity	Sophomore-Spring	0.213
Semester Workload Intensity	Junior-Spring	0.496
Semester Failure Rate	Sophomore-Fall	0.077
Semester Failure Rate	Junior-Fall	0.391
Semester Failure Rate	Junior-Spring	0.772
Semester Weekly Hours	Sophomore-Fall	0.147
Semester Weekly Hours	Sophomore-Spring	0.123
Semester Weekly Hours	Junior-Fall	0.545
Semester Weekly Hours	Junior-Spring	0.789

Twenty-three of 28 treatment-level cells fail to reject normality. Cells that *do* reject normality are reported separately in Table 2.

2. *Default Layout*: flowchart shown, then ratings for Difficulty (0–100), Workload Intensity / Burnout (0–100), Anticipated Graduation Rate (0–100).
3. *Layout A* (DSM $\leftrightarrow$ PSM): same three ratings.
4. *Layout B* (AES, DDDM I back one semester): same three ratings.
5. *Layout C* (combination of A and B): same three ratings.
6. *Cross-layout preference*: Preference rating (0–100) for each of Default, A, B, C.
7. *Cross-layout graduation rate*: Anticipated graduation rate (0–100) for each of Default, A, B, C, presented side-by-side.
8. *Engagement (humor)*: “If ISE had a mascot, what would it be and why?”; “Which professor would you want to get drinks with?”; “If possible, which professor would you like to see do the worm at the ISE Banquet?” (free text per [Krosnick \(1991\)](#); [McDaniel and Gates \(2018\)](#)).
9. *Default-layout semester ratings*: Workload intensity (0–100), anticipated failure percentage (0–100), and weekly hours worked (0–40), each repeated for Sophomore-Fall, Sophomore-Spring, Junior-Fall, Junior-Spring.
10. *External factors*: “What are some external factors that have affected your progress toward your degree?” (free text).